

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 1 (Generated practice paper - Set 13)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

**Answer all questions in the spaces provided.**

**0 1**

The Universal Soil Loss Equation (USLE) is  $A = R \times K \times LS \times C \times P$ . Table 1 gives factor values for a farm.

**Table 1**

| Factor                  | Description                                                                                         | Value        |
|-------------------------|-----------------------------------------------------------------------------------------------------|--------------|
| R                       | Rainfall erosivity for the farm                                                                     | 110          |
| K                       | Soil erodibility: loam                                                                              | 0.67         |
| LS                      | Slope length and gradient: 120 m, 6%                                                                | 1.35         |
| C (crop type x tillage) | Beans (0.50) with spring ploughing (0.90); or wheat (0.35) with mulch tillage (0.60)                | see question |
| P                       | Ploughing up and down slope 1.0; contour ploughing 0.50; strip cropping with contour ploughing 0.25 | see question |

**Table 2**

| Soil erosion class | Rate of soil loss / $\text{t ha}^{-1} \text{ yr}^{-1}$ |
|--------------------|--------------------------------------------------------|
| Low                | under 5                                                |
| Moderate           | 5-10                                                   |
| High               | 10-20                                                  |
| Very high          | 20-30                                                  |
| Severe             | over 30                                                |

**0 1**

**1**

Calculate the annual soil loss for beans with spring ploughing, ploughed up and down the slope. Give your answer to one decimal place.

**[2 marks]**

**Answer**

t/ha/yr

**0 1**

**2**

Calculate the annual soil loss for wheat with mulch tillage and strip cropping with contour ploughing.

**[2 marks]**

Answer  t/ha/yr

**0 1 . 3** The natural rate of soil formation on the farm is  $1 \text{ t ha}^{-1} \text{ yr}^{-1}$ . Use your answers and Table 2 to evaluate the sustainability of the two farming systems. **[3 marks]**

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**0 1 . 4** Explain why the USLE gives only an estimate of soil loss. **[2 marks]**

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**0 1 . 5** Suggest which single factor in the USLE a farmer can change most easily. **[1 mark]**

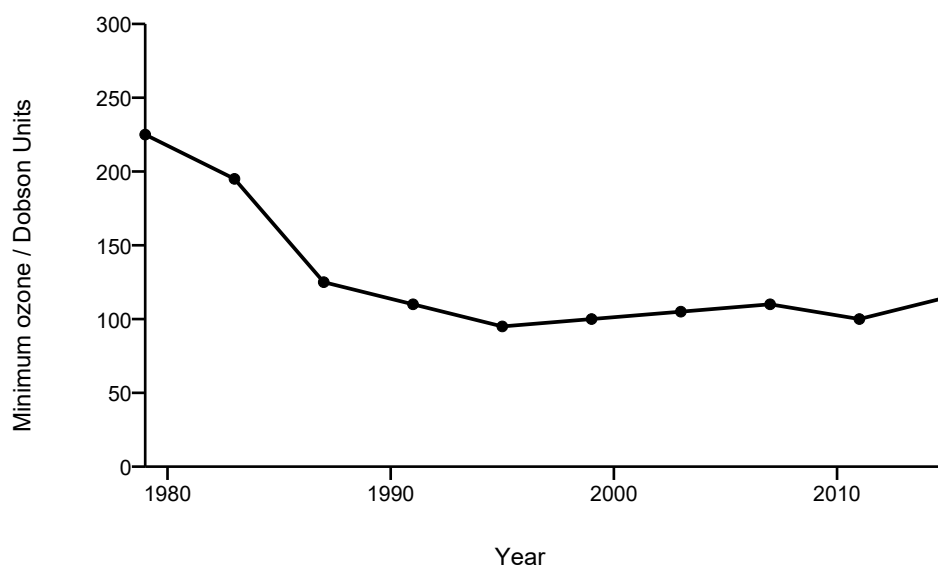
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**10**

**0 2**

Figure 1 shows the minimum ozone concentration recorded over Antarctica each spring between 1979 and 2015.

**Figure 1****0 2****1**

Describe the trend shown in Figure 1.

**[2 marks]****0 2****2**

Calculate the percentage decrease in minimum ozone between 1979 and 1995.

**[2 marks]****Answer**

**%****0 2****3**

Explain why ozone depletion is most severe over Antarctica in spring.

**[4 marks]**

0 2 . 4

Explain why the data in Figure 1 alone could not confirm that CFCs were the cause of the depletion, and describe how the cause was confirmed.

[2 marks]

10

0 3 . 1

Explain why nitrates in drinking water can cause **blue baby syndrome** (methaemoglobinaemia).

[3 marks]

0 3 . 2

Explain why the link between nitrates in drinking water and stomach cancer is uncertain.

[2 marks]

0 3 . 3

Describe **three** ways farmers in a Nitrate Vulnerable Zone can reduce nitrate leaching into aquifers.

[3 marks]

1 .....

2 .....

3 .....

0 3 . 4

Explain how phosphates are removed from sewage effluent and from lakes polluted in the past.

[2 marks]

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0 4 . 1

Describe how bioleaching is used to extract copper from low-grade ores or spoil heaps, and explain why it may be more sustainable than smelting.

[5 marks]

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04 . 2

Describe the process of phytomining.

[3 marks]

04 . 3

Explain how polymer adsorption may provide a low-energy source of uranium in the future.

[2 marks]

10

05 . 1

Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

**Emission location** .....

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**Emission timing** .....

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**0 5 . 2**

Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

**[2 marks]**

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**0 5 . 3**

Explain the **precautionary principle** as applied to the release of a new chemical.

**[2 marks]**

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**10**

06

A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

06

. 1

Calculate the estimated population size.

[2 marks]

Answer

beetles

06

. 2

State **four** assumptions of the Lincoln Index method.

[4 marks]

1

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. 3

Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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06

. 4

Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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10

07 Mercury released into a lake from a factory was in the inorganic form. Fish in the lake were later found to contain high concentrations of methyl mercury, an organic form.

07 . 1 Suggest how the inorganic mercury was converted to methyl mercury.  
[2 marks]

07 . 2 Explain why methyl mercury is a more hazardous pollutant than inorganic mercury.  
[3 marks]

07 . 3 Explain why the health risk from eating fish from the lake may remain for many years after the discharges stop.  
[2 marks]

07 . 4 Suggest **three** factors that should be included in a Critical Pathway Analysis of a proposed mercury discharge.  
[3 marks]

1 .....

2 .....

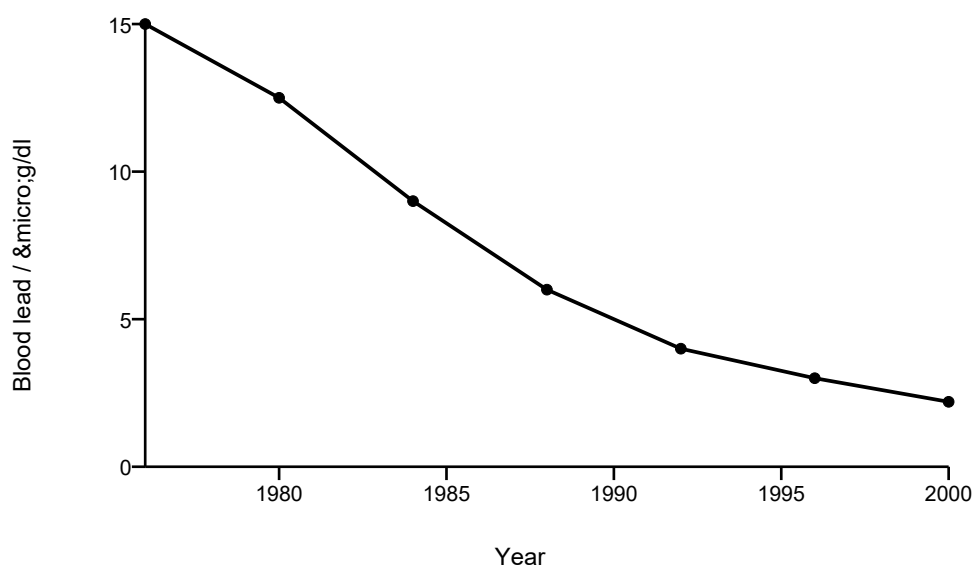
3 .....

10

0 8

Figure 1 shows the mean concentration of lead in the blood of children in a large city between 1976 and 2000, and the year in which lead additives in petrol began to be phased out.

Figure 1



0 8

1 Calculate the percentage decrease in mean blood lead between 1976 and 2000.

[2 marks]

Answer

%

08 . 2

Explain why tetraethyl lead was added to petrol and why this caused a public health threat.

[3 marks]

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08 . 3

Suggest **two** reasons, other than petrol additives, why blood lead concentrations continued to fall after 1990.

[2 marks]

1 .....

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2 .....

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08 . 4

Explain how adding small amounts of phosphoric acid to a water supply reduces lead in drinking water.

[2 marks]

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08 . 5

Explain why discarded lead fishing weights and shotgun pellets can poison waterfowl.

[1 mark]

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09 . 1

Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

- A High liposolubility and high persistence
- B Low liposolubility and low persistence
- C High liposolubility and low persistence
- D Low liposolubility and high persistence

☐☐☐☐

09 . 2

Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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09 . 3

Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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10 . 1

Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

[9 marks]

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[illegible]

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**10** . **2** State what is meant by **peak shaving**.

**[1 mark]**

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**10**

Do not write outside the box

**Write an essay on one of the following topics.**

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | . | 1 |
|---|---|---|---|

Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

**[25 marks]**

**OR**

**1 1 . 2**

Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.



[illegible]



**END OF QUESTIONS**